

Physical Chemistry of Materials

Lecture 03

Phase Transitions



Equilibrium

- An isolated system is **in equilibrium** when its properties remain **constant with time**
- Mechanical equilibrium: constant p and V
- Diffusive equilibrium: constant n
- Thermal equilibrium: constant T



Last Class

First Law of Thermodynamics

“The internal energy of an isolated system is constant.”

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$dU = \delta q + \delta w$$

↑ ↑ ↑
small change of U small q small w



Entropy

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Index of exhaustion
- Second law of thermodynamics

$$dS \geq 0 \text{ (isolated system)}$$

- $dS = 0$: reversible change
- $dS > 0$: irreversible change
- Third law of thermodynamics: $\lim_{T \rightarrow 0} S = 0$



Fundamental Equation

From the first law,

$$dU = \delta q + \delta w$$

For a reversible change,

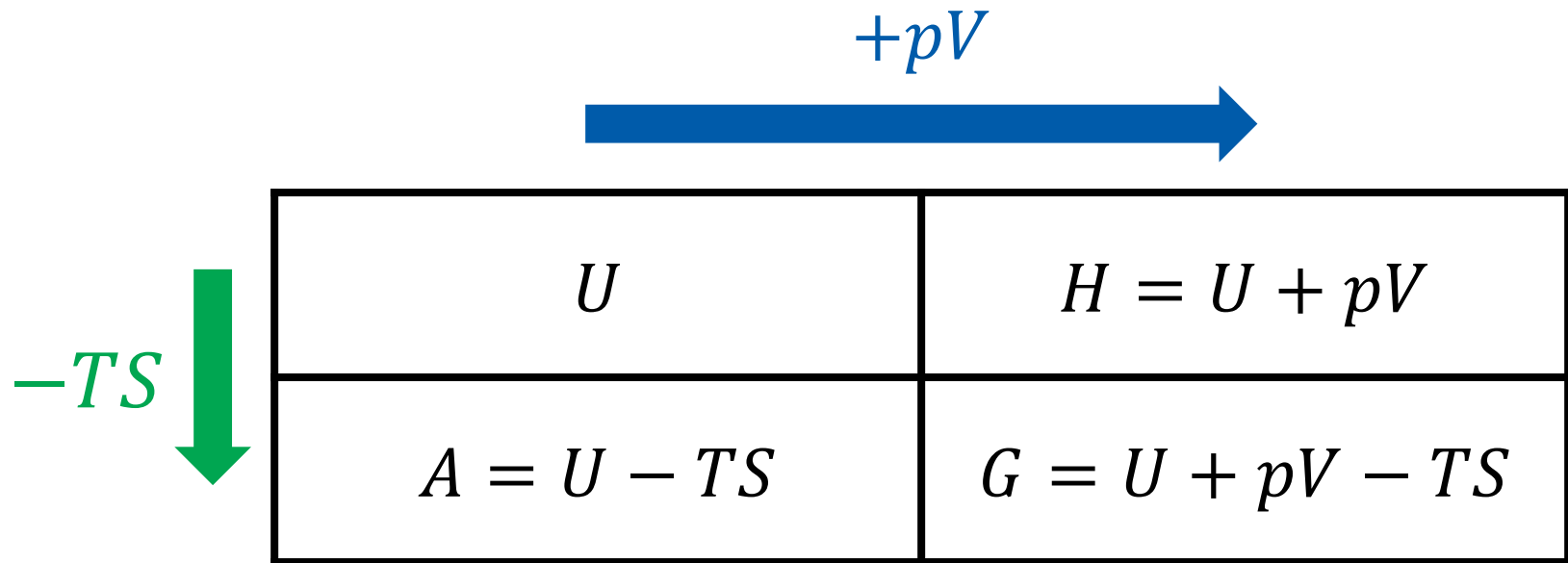
$$dU = \delta q_{\text{rev}} + \delta w_{\text{rev}} = TdS - pdV$$

Since all variables are state functions,

$$dU = TdS - pdV \text{ (in general)}$$

Last Class

Thermodynamic Potentials



Two Systems in Equilibrium

$$dU = TdS - pdV + \mu dn$$

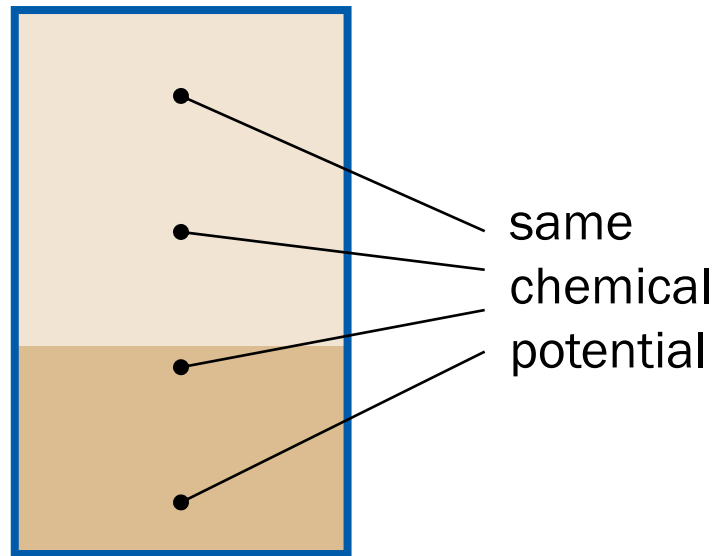
- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn



Last Class

Thermodynamics of Phases

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system

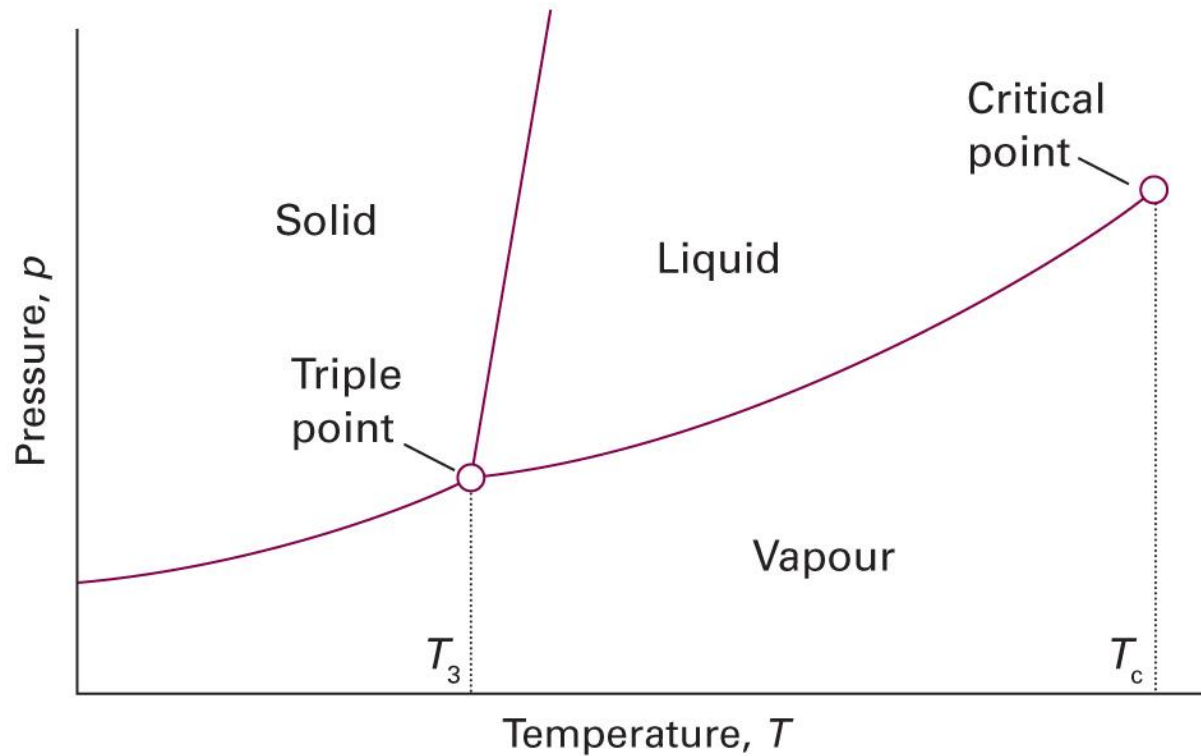


Phase

A form of matter that is **uniform** throughout in **chemical composition** and **physical state**

- **Phase diagram:** A diagram that shows the regions where its various phases are thermodynamically stable

Typical Phase Diagram

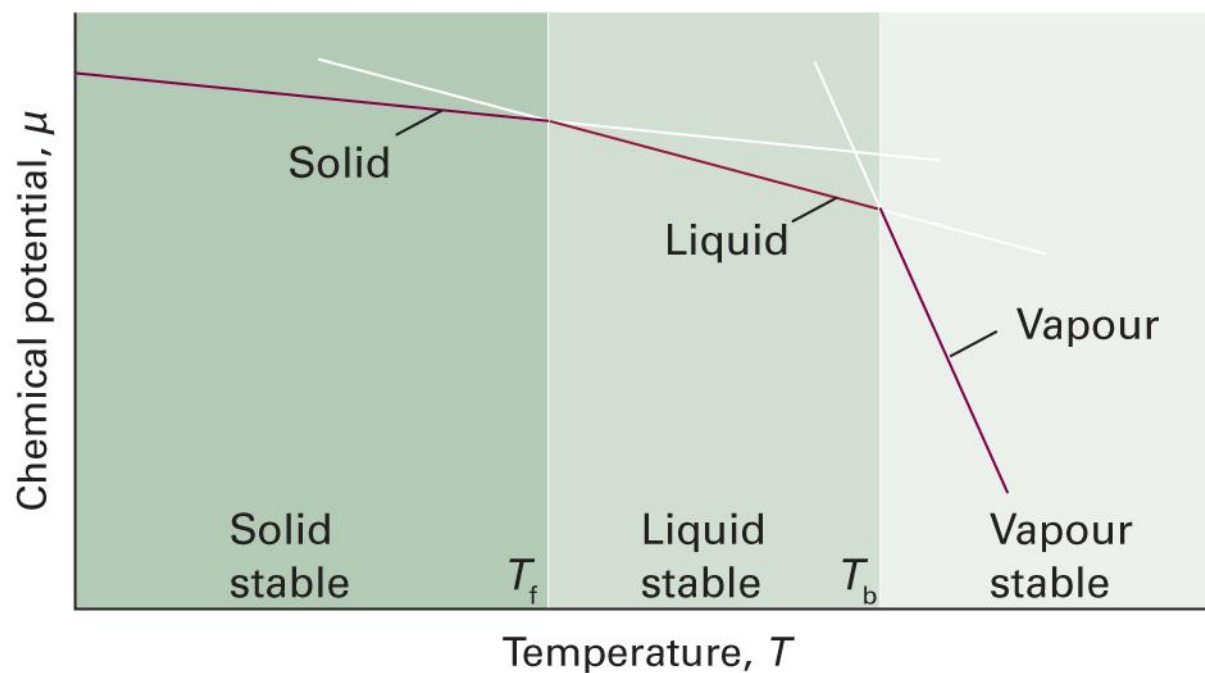


Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Chemical Potential

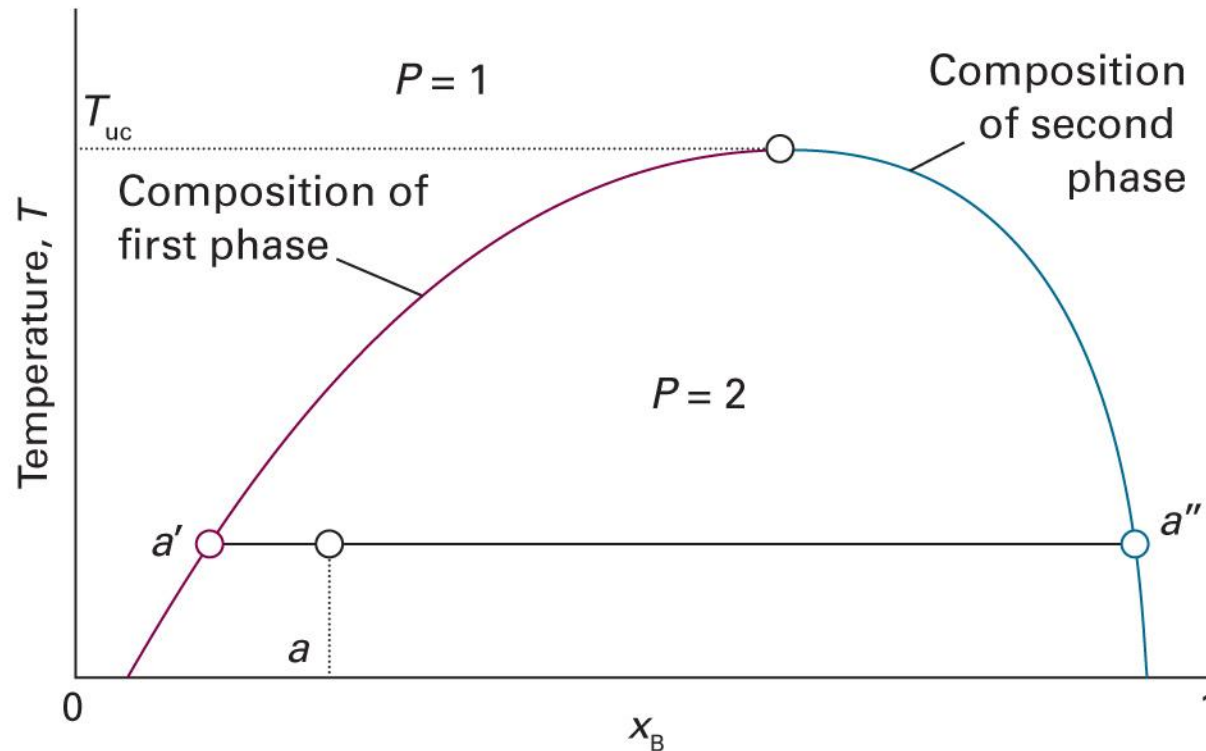


(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 4B.1)

Multi-Component System

- Possible phases
 - Each component: gas, liquid, solid
 - Mixed/demixed
- e.g. two-component system (binary system)
 - A: liquid, B: liquid, A-B: mixed
 - A: liquid, B: liquid, A-B: demixed
 - A: gas, B: gas, A-B: mixed
 - ...

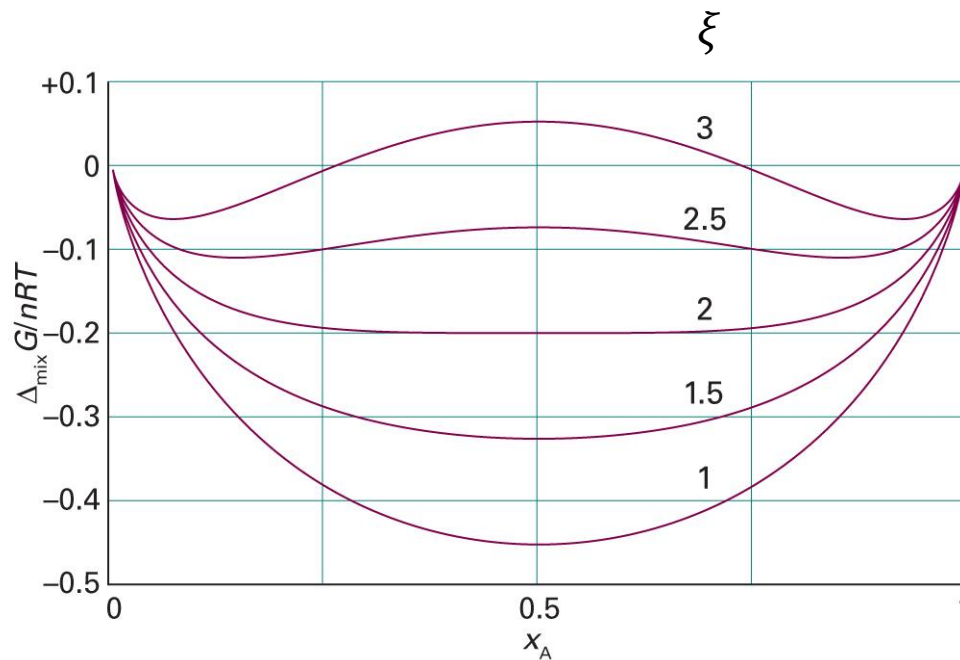
Example: Binary System



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5C.14)

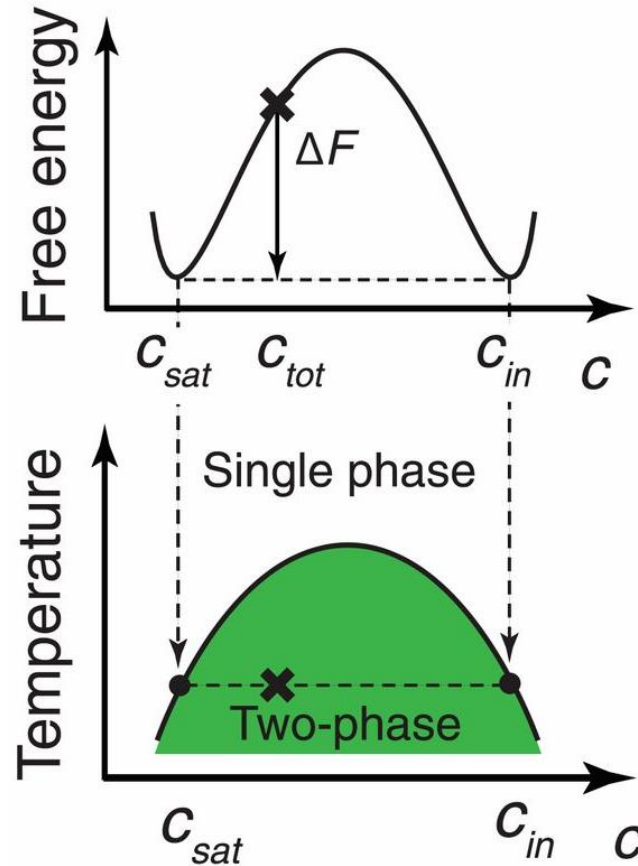
Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



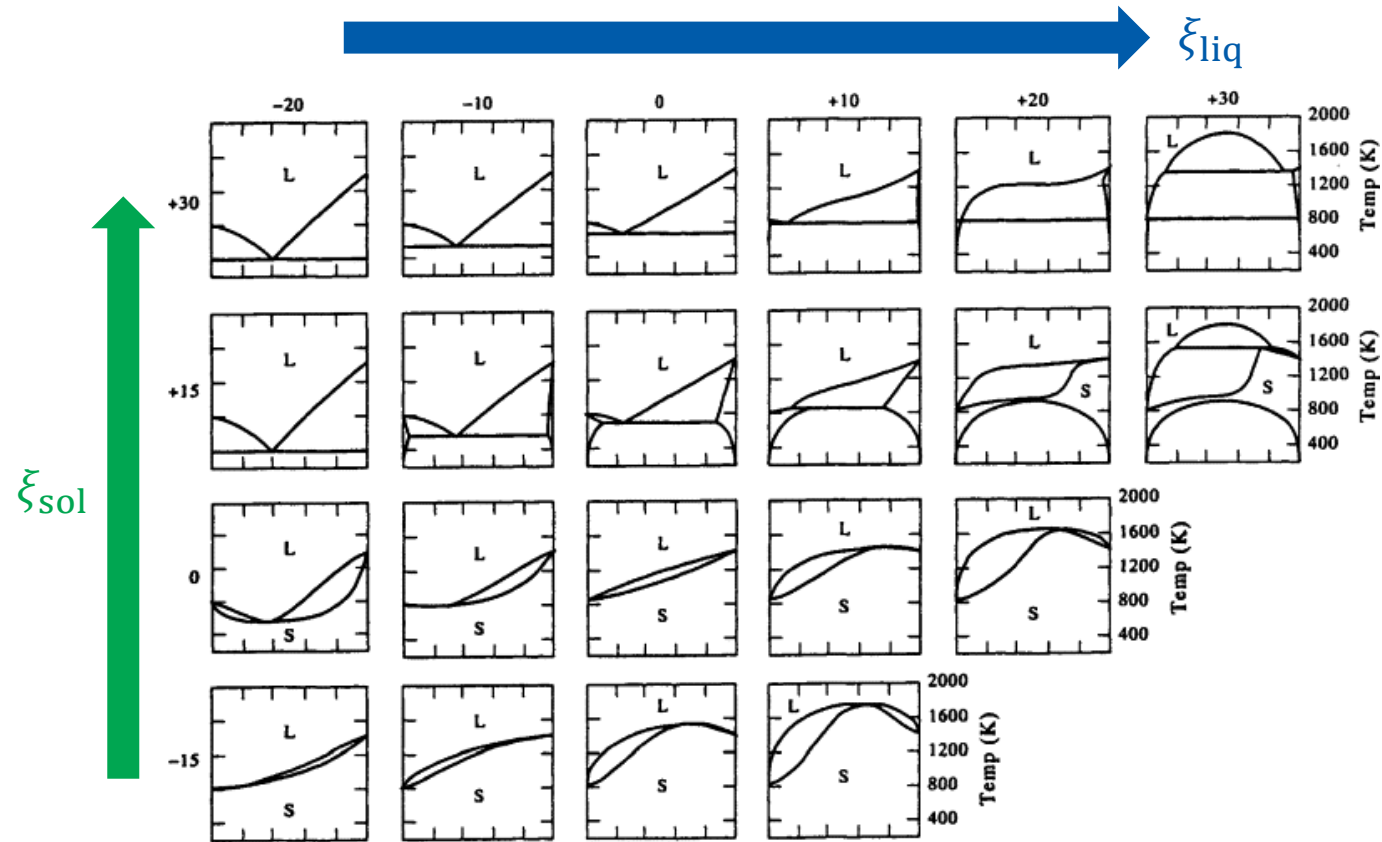
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

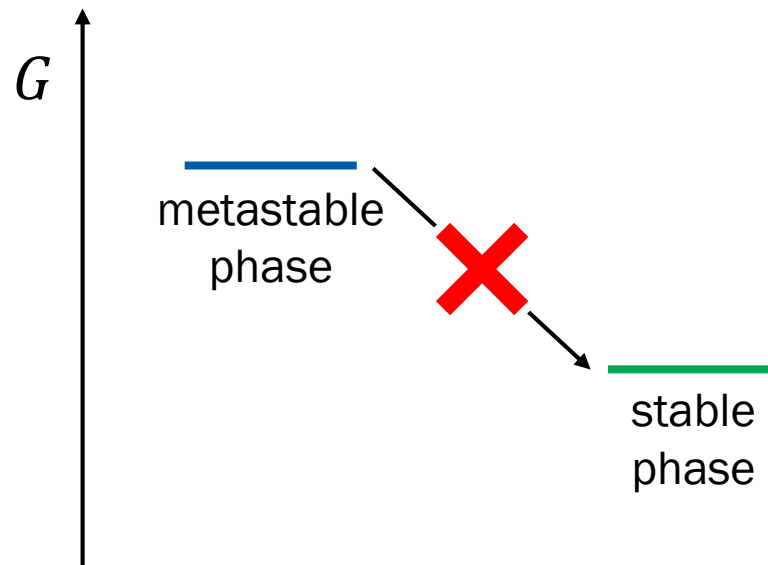
Liquid-Solid Binary Mixture



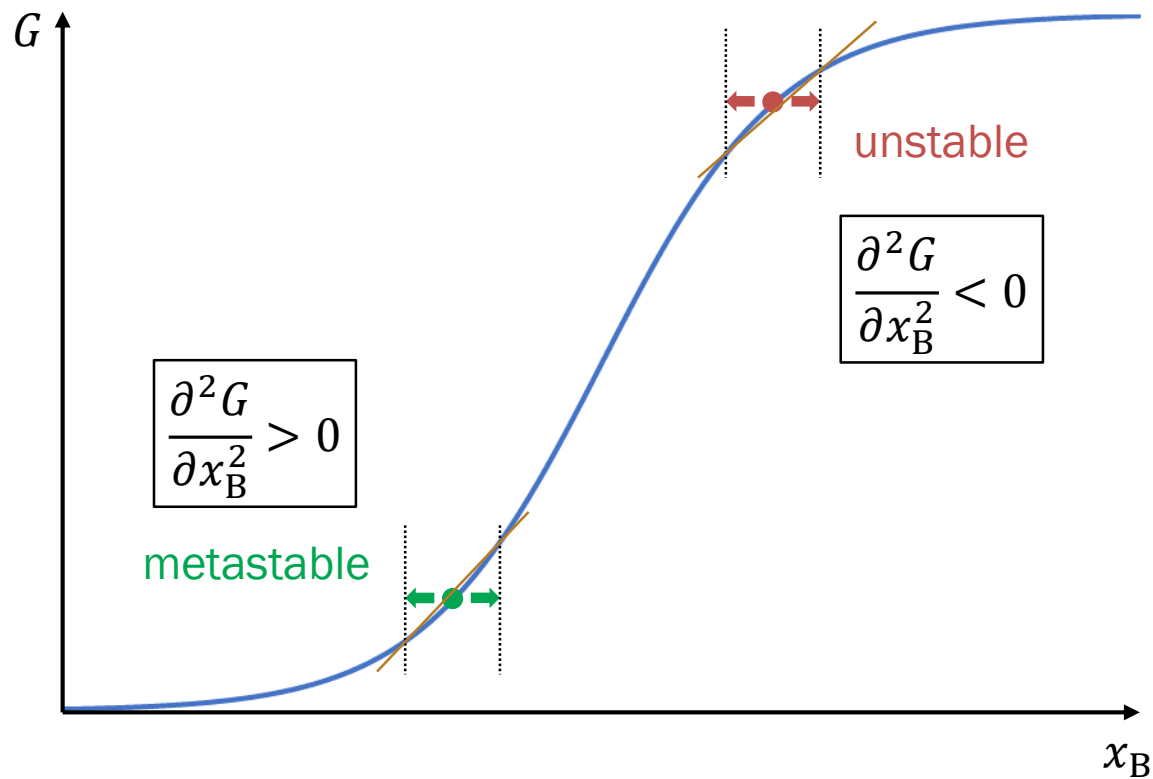
(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

Metastable Phase

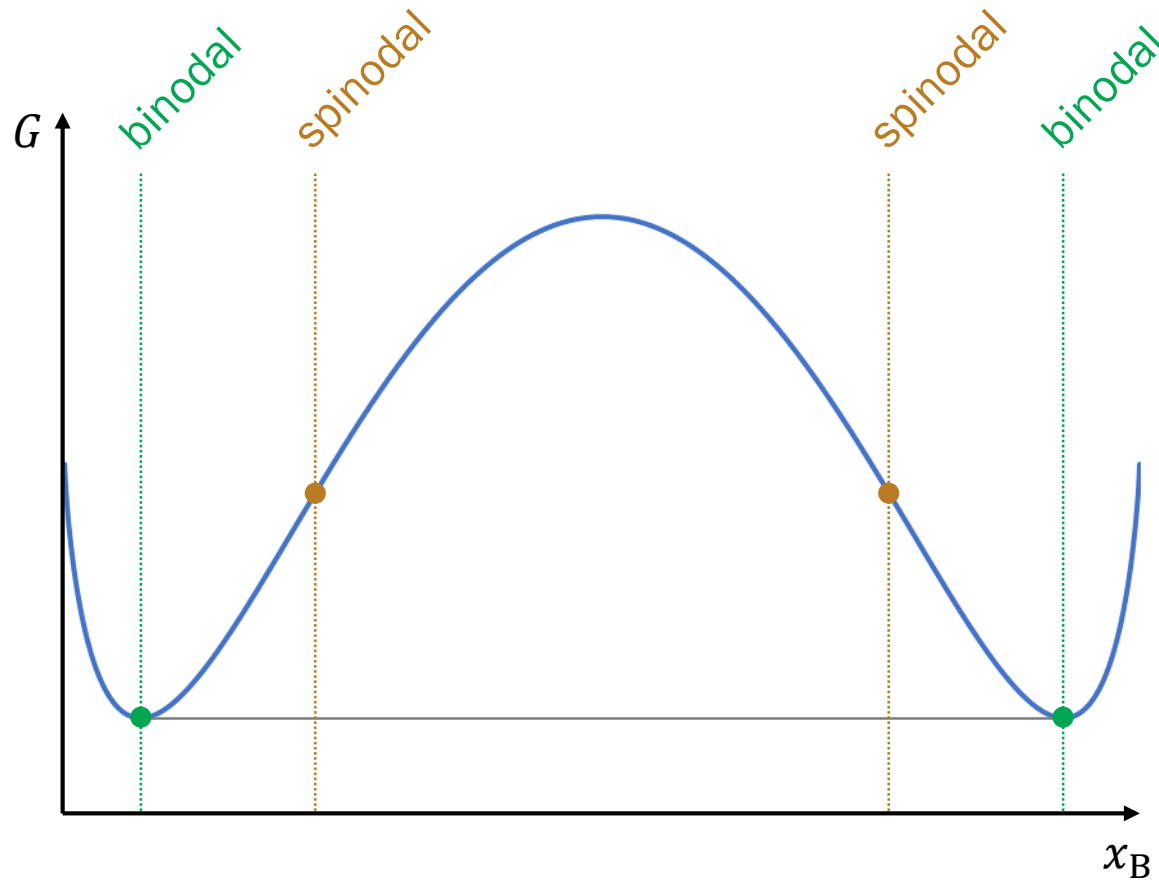
Thermodynamically unstable phases that persist because the transition is kinetically hindered



Metastable? Unstable?



Binodal and Spinodal



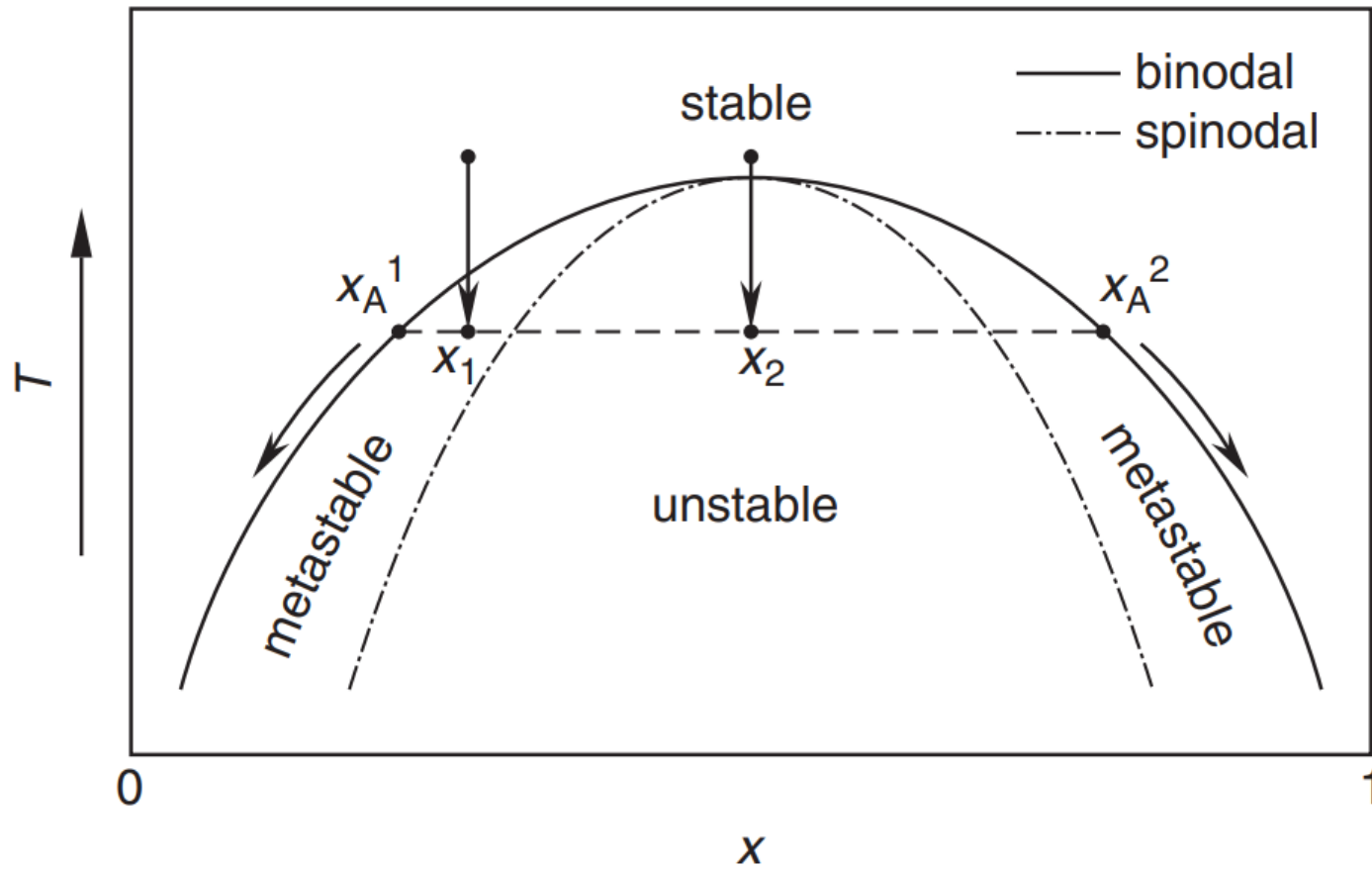
Binodal:

$$\frac{\partial G}{\partial x_B} = 0$$

Spinodal:

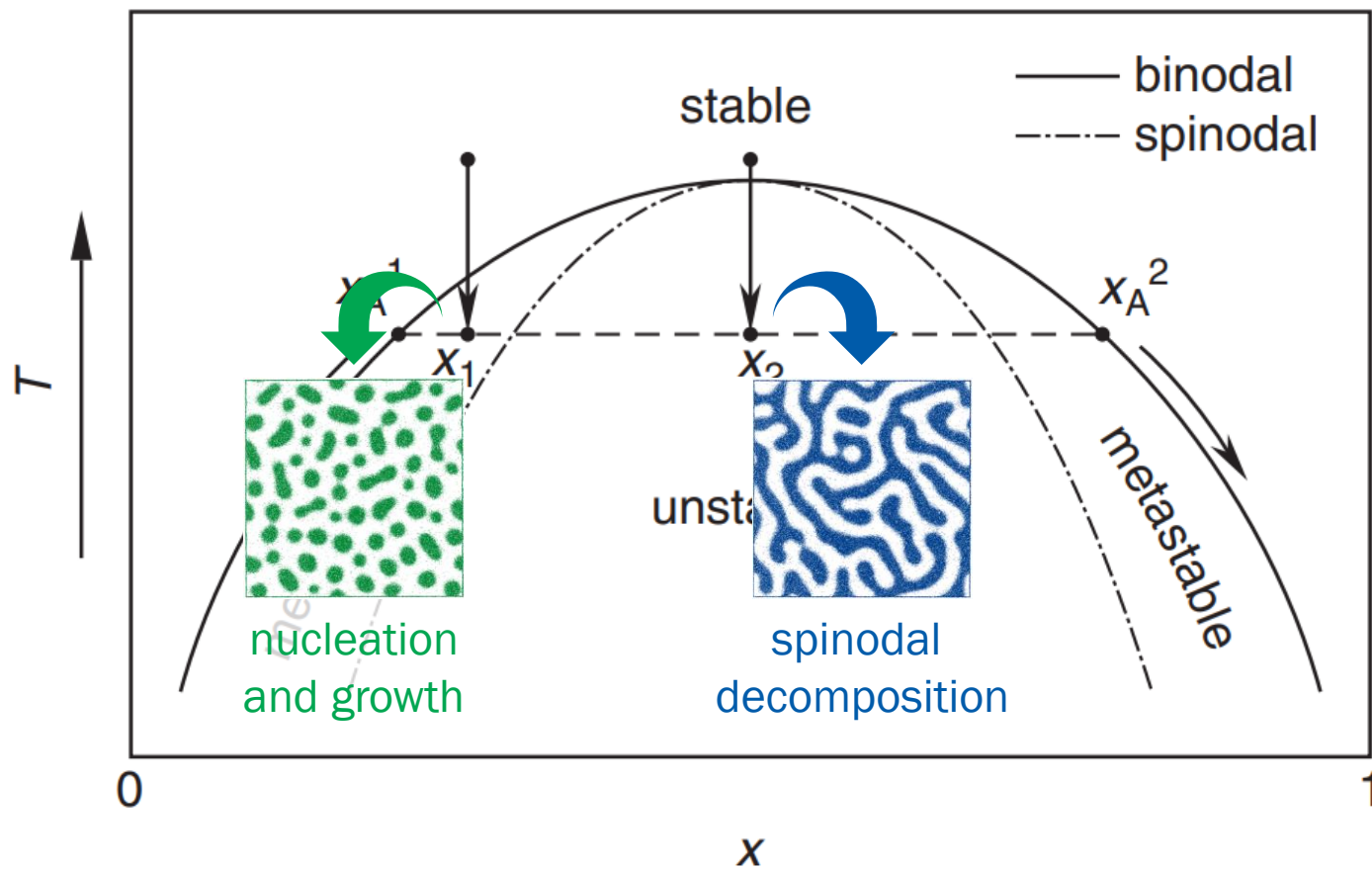
$$\frac{\partial^2 G}{\partial x_B^2} = 0$$

Two Lines on Phase Diagram



(Image: Cheng, *Phase Transitions in Polymers* (2008), Fig. 2.14)

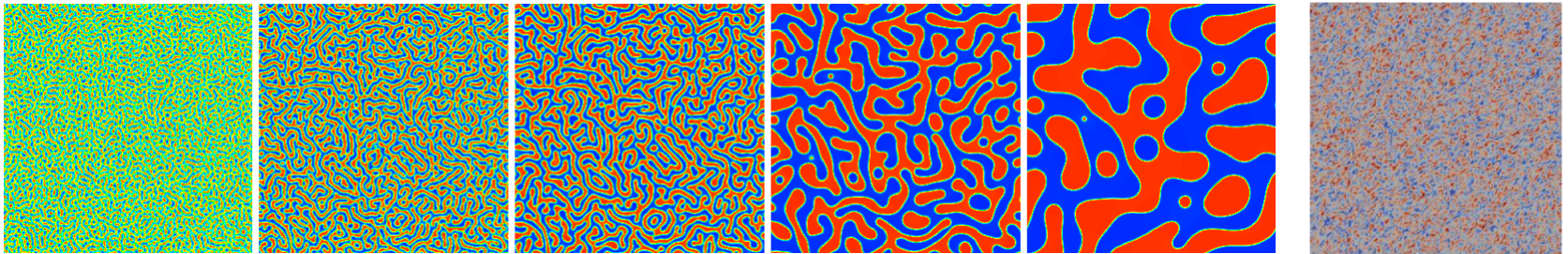
Kinetics of Phase Separation



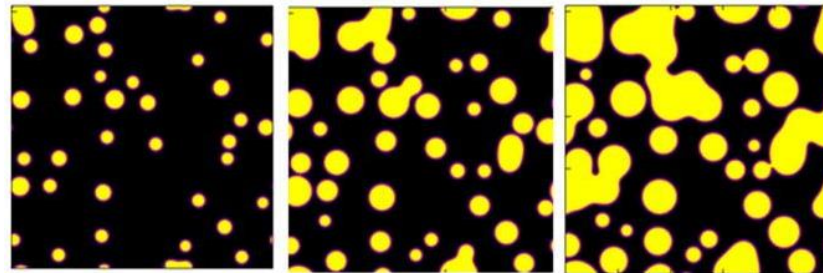
(Images: Cheng, *Phase Transitions in Polymers* (2008), Fig. 2.14 / Gebauer et al., *Chem Soc Rev* (2014), Fig. 3) 22

Kinetics of Phase Separation

- Spinodal decomposition



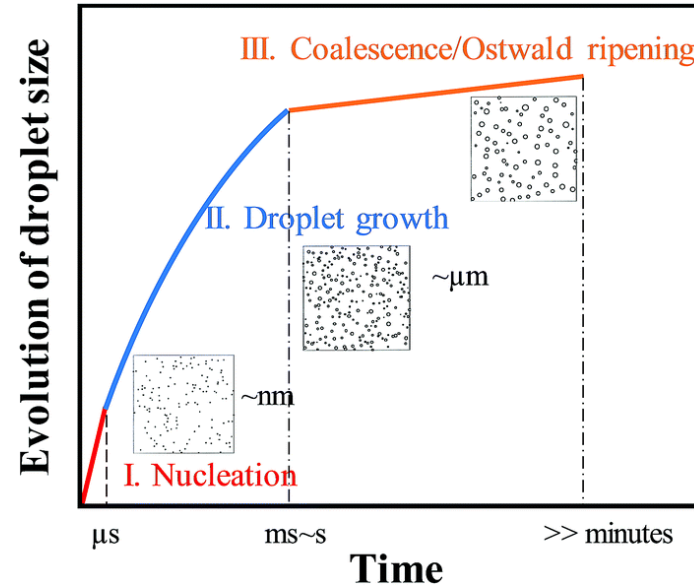
- Nucleation and growth 



(Images: <https://doi.org/10.2316/P.2013.804-040>
<https://fuckphasefield.com/spinodal-decomposition-in-2d/>
<https://doi.org/10.1007/s11837-014-1033-9>)

Nucleation and Growth

metastable phase → stable phase



Universal in chemistry and physics!

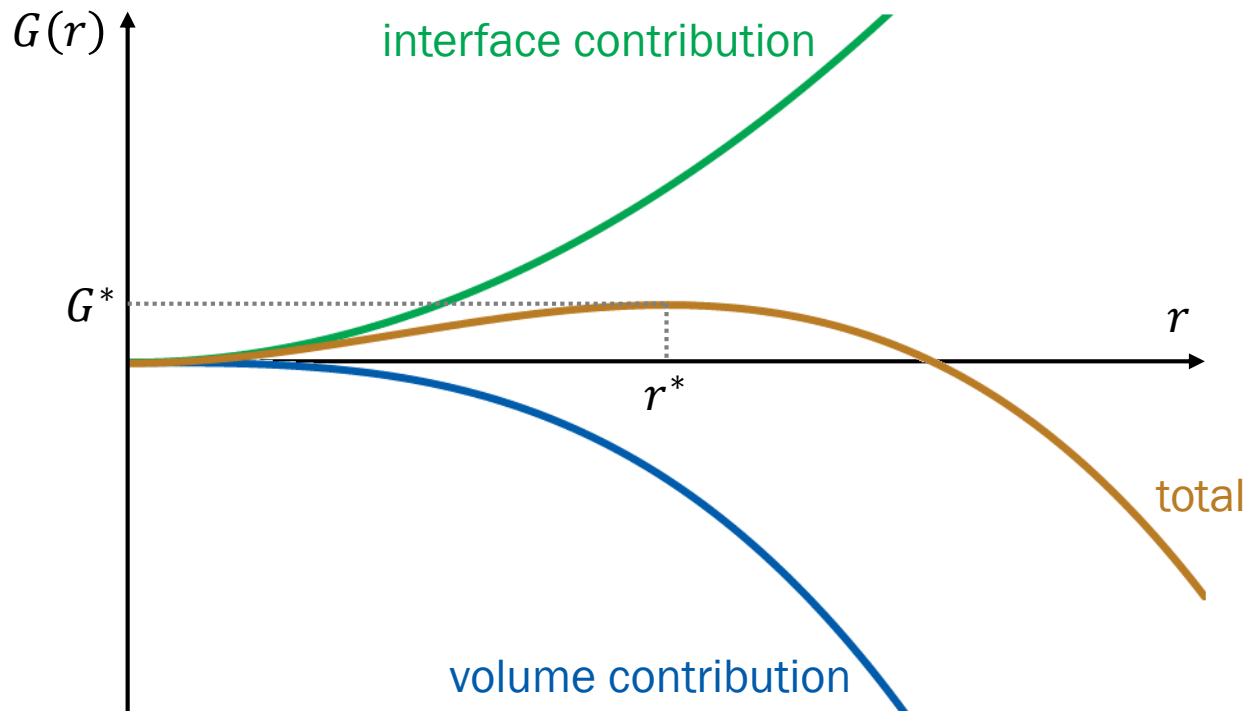
(Image: Luo et al., *Phys Chem Chem Phys* (2017), Fig. 6)

Classical Nucleation Theory

- Gibbs energy density $\Delta g = g_{\text{unstable}} - g_{\text{stable}} < 0$
- Surface tension $\gamma > 0$
- Spherical droplet of size r
- Gibbs energy of the droplet

$$G(r) = V\Delta g + A\gamma = \frac{4}{3}\pi r^3 \Delta g + 4\pi r^2 \gamma$$

Gibbs Energy of Droplet



Critical Size

$$G(r) = \frac{4}{3}\pi r^3 \Delta g + 4\pi r^2 \gamma$$

- Critical size r^*

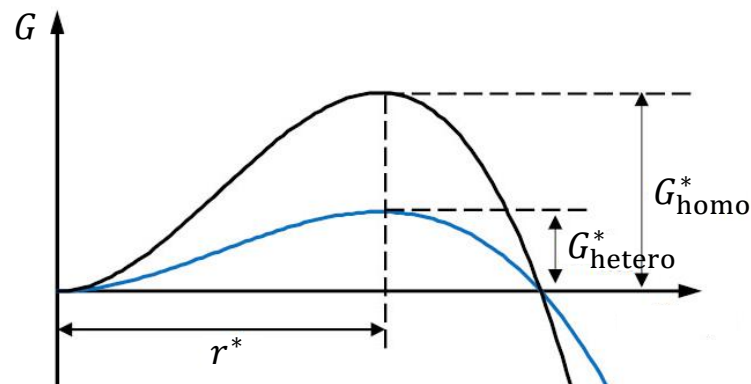
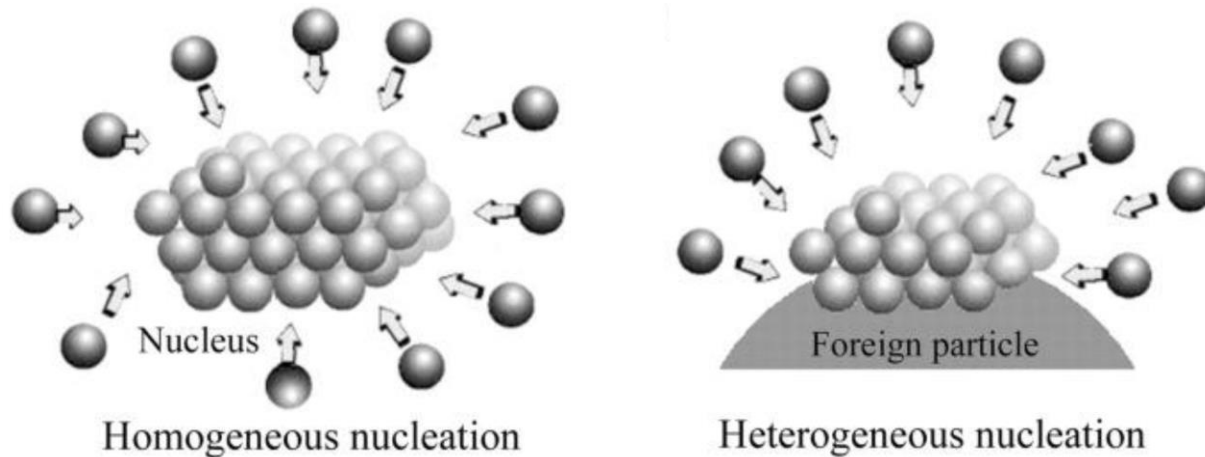
$$\frac{dG(r)}{dr} = 4\pi r^2 \Delta g + 8\pi r \gamma = 0$$

$$r^* = -2\gamma/\Delta g \text{ (nontrivial)}$$

- Activation energy G^*

$$G^* = G(r^*) = \frac{4}{3}\pi \left(-\frac{2\gamma}{\Delta g}\right)^3 \Delta g + 4\pi \left(-\frac{2\gamma}{\Delta g}\right)^2 \gamma = \frac{16\pi\gamma^3}{3(\Delta g)^2}$$

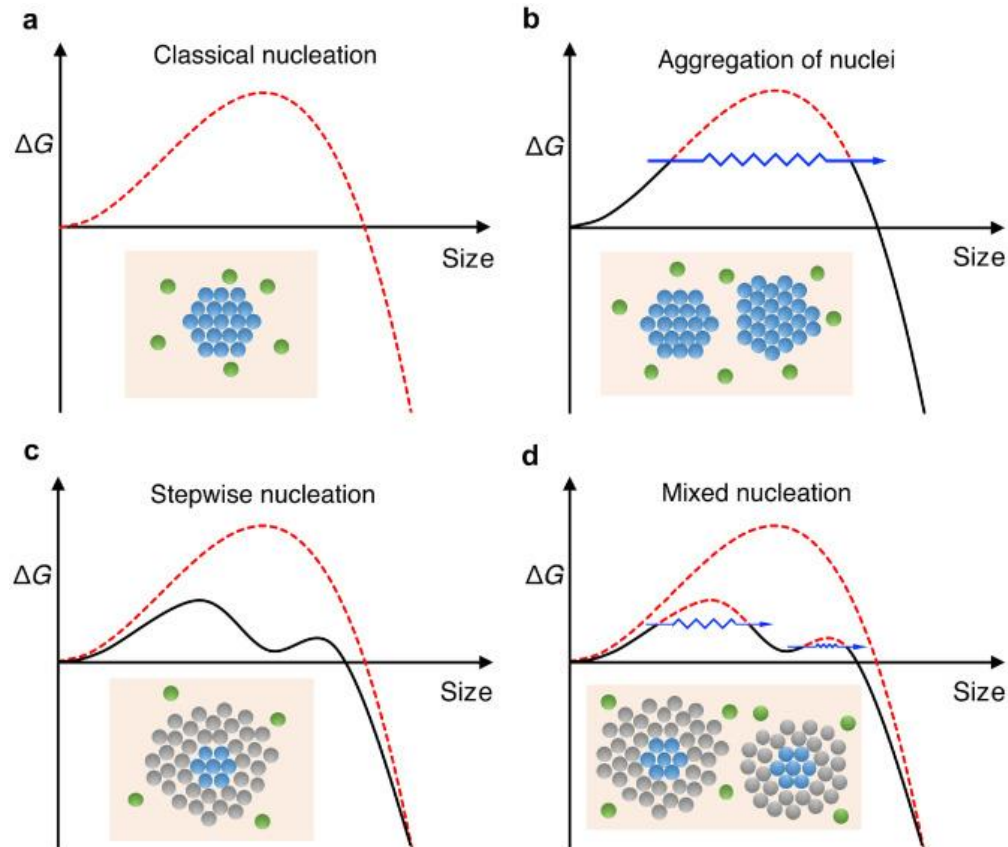
Homogeneous vs. Heterogeneous



(Images: Chen et al., *Sustainability* (2022), Fig. 7 /

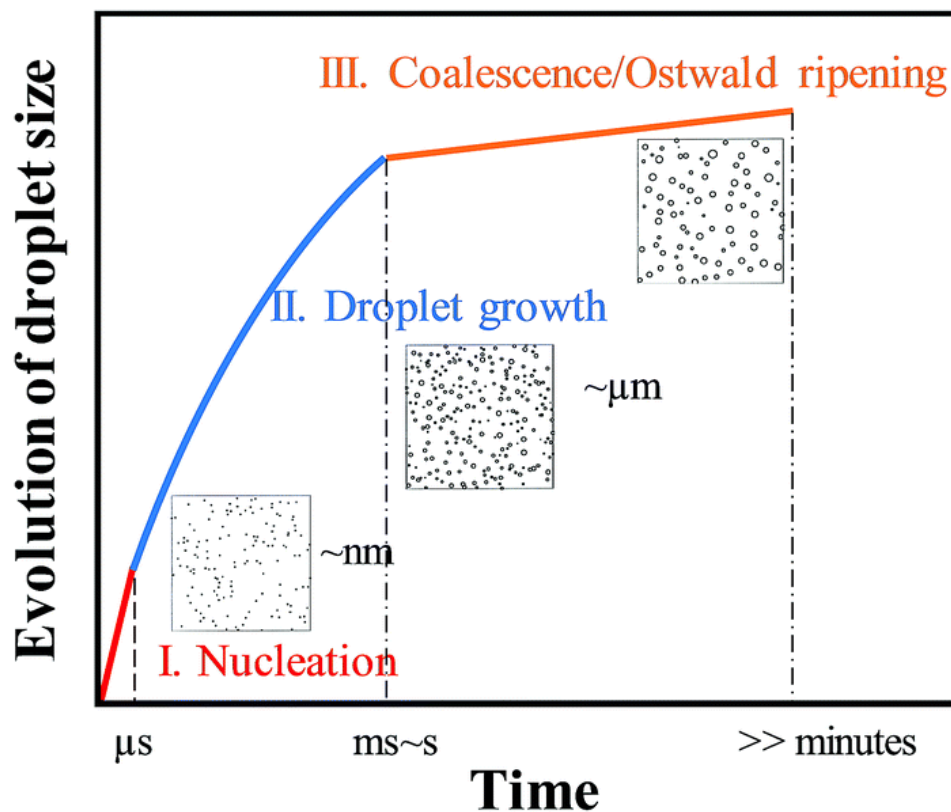
<https://www.elementum3d.com/white-papers/understanding-solidification-and-nucleation-in-aluminum-alloys/>)

Nonclassical Theories



(Image: Ou et al., *Acta Materialia* (2022), Fig. 1)

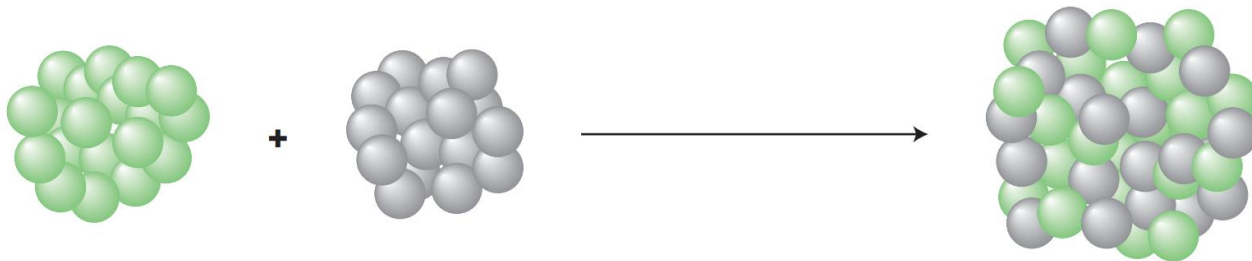
Nucleation and Growth



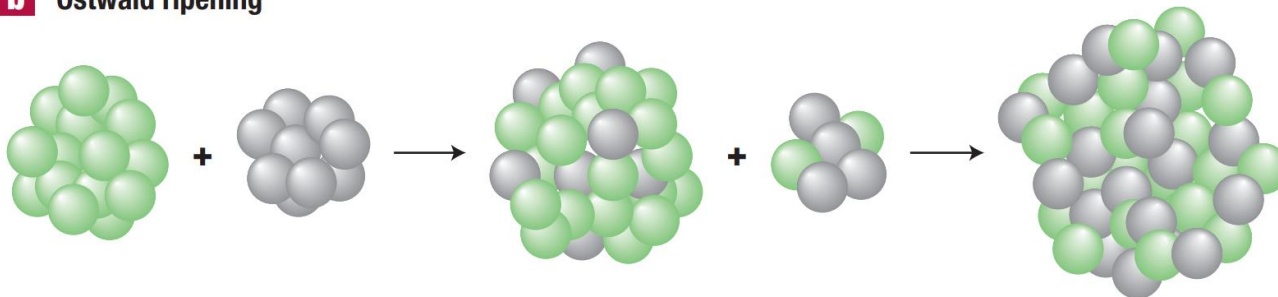
(Image: Luo et al., *Phys Chem Chem Phys* (2017), Fig. 6)

Late Stage of Growth

a Coalescence

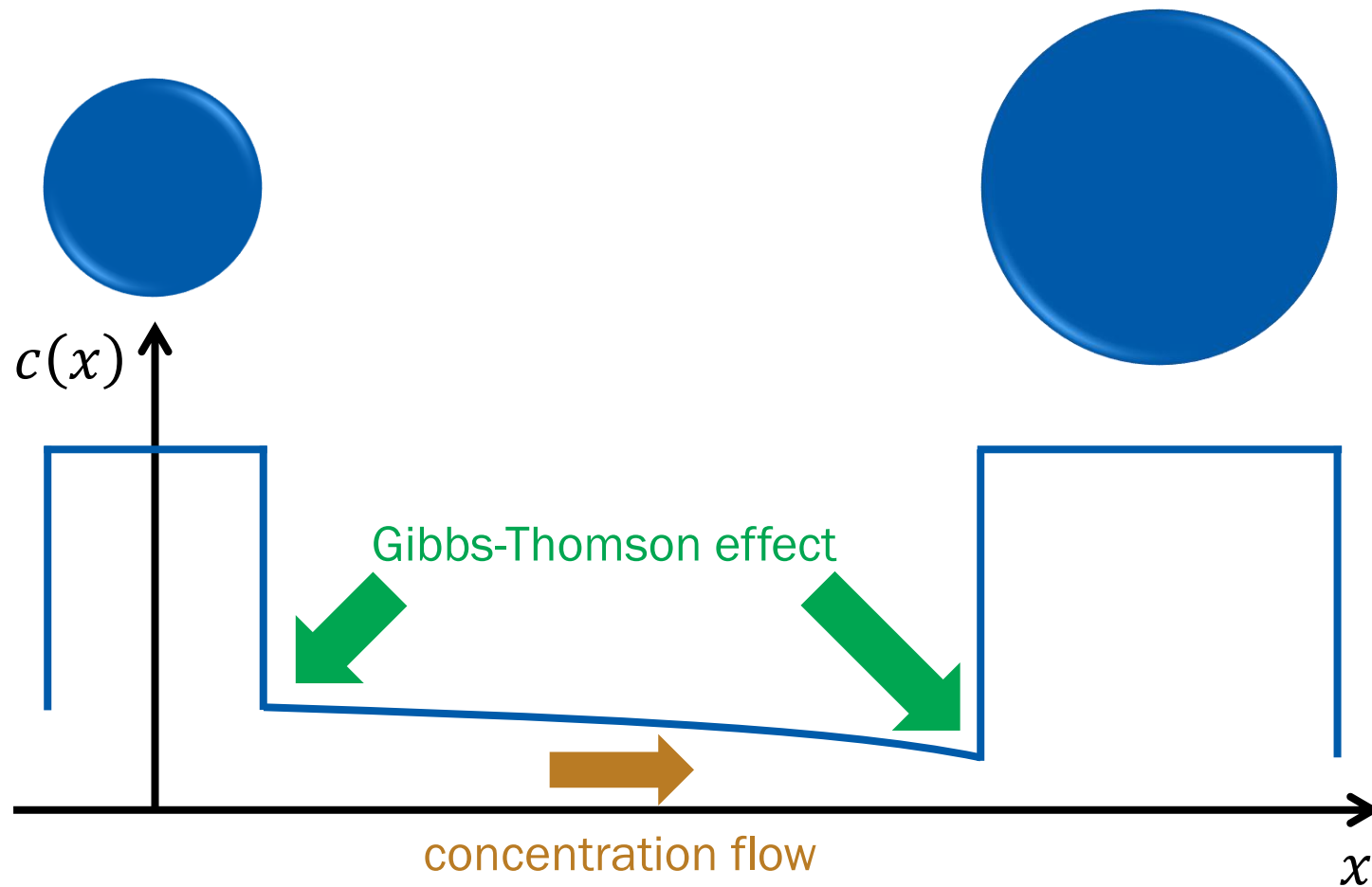


b Ostwald ripening



(Image: Bowker, *Nature Materials* (2002), Fig. 1)

Ostwald Ripening



Thermodynamics of Droplet

- Interface effect: surface tension

$$dU = TdS - pdV + \gamma d\sigma + \mu dn$$

- Spherical droplet of size r
 - Volume $V = (4/3)\pi r^3$
 - Mole number $n = V/V_m = (4\pi r^3)/(3V_m)$
 - Surface area $\sigma = 4\pi r^2$

Hence,

$$dn = (4\pi r^2/V_m)dr$$

$$d\sigma = 8\pi r dr = (2V_m/r)dn$$



Chemical Potential of Droplet

$$dU = TdS - pdV + \gamma d\sigma + \mu dn$$

$$dn = (4\pi r^2/V_m)dr$$

$$d\sigma = 8\pi r dr = (2V_m/r)dn$$

Substitution leads to

$$dU = TdS - pdV + \left(\mu + \frac{2\gamma V_m}{r} \right) dn$$

$$\mu_{\text{droplet}} = \mu_{\text{bulk}} + \frac{2\gamma V_m}{r}$$

Diffusive Equilibrium

$$\mu_{\text{droplet}} = \mu_{\text{bulk}} + \frac{2\gamma V_{\text{m}}}{r}$$

Chemical potentials must be identical:

$$\mu_{\text{solution}} = \mu_{\text{droplet}}$$

$$\mu_{\text{solution}} = \mu^{\ominus} + RT \ln a \approx \mu^{\ominus} + RT \ln c$$

Hence,

$$\mu^{\ominus} + RT \ln c = \mu_{\text{bulk}} + \frac{2\gamma V_{\text{m}}}{r}$$

Gibbs-Thomson Effect

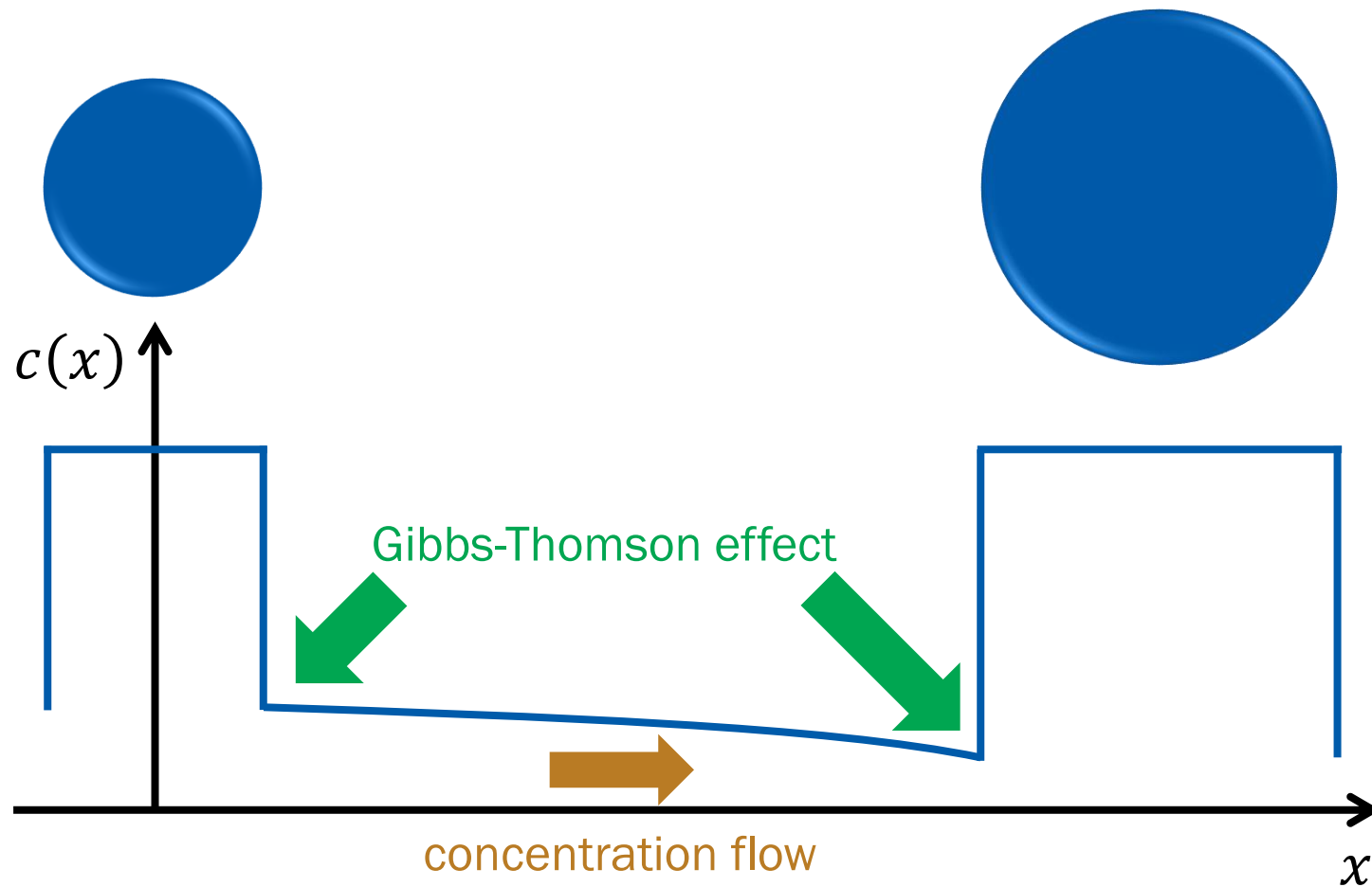
$$\mu^\ominus + RT \ln c = \mu_{\text{bulk}} + \frac{2\gamma V_{\text{m}}}{r}$$

$$c(r) = A \exp\left(\frac{2\gamma V_{\text{m}}}{rRT}\right)$$

As $c(\infty) = A$,

$$c(r) = c(\infty) \exp\left(\frac{2\gamma V_{\text{m}}}{rRT}\right)$$

Ostwald Ripening

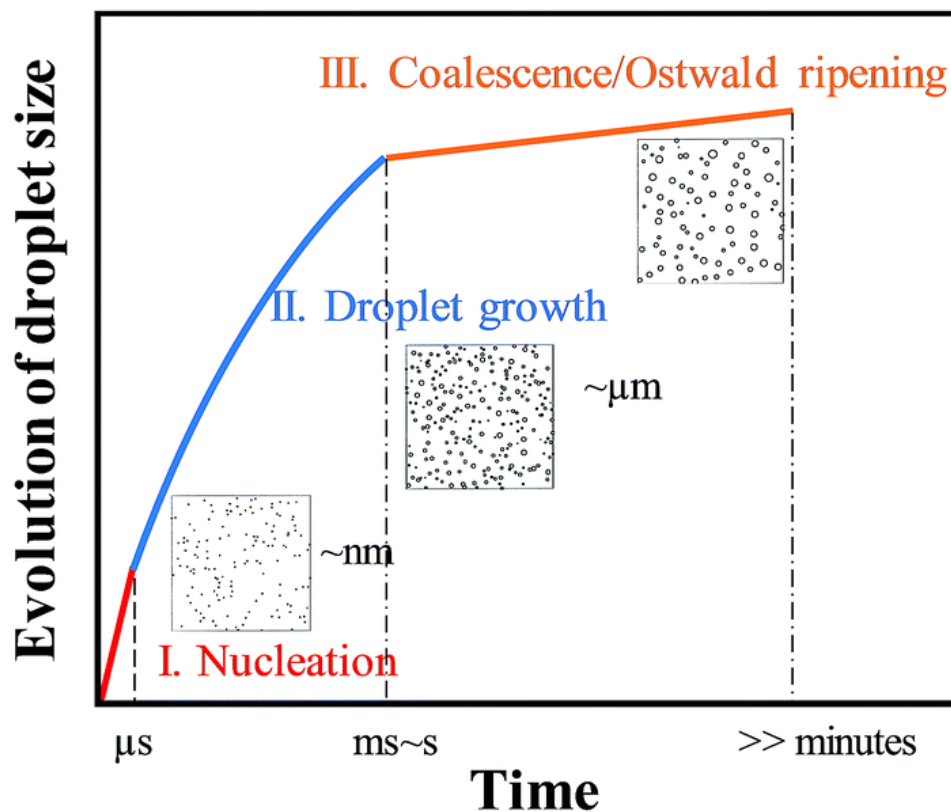


Ostwald Ripening



(Image: Huang *et al.*, *Nature Communications* (2017), Supp. Movie 1)

Nucleation and Growth



(Image: Luo et al., *Phys Chem Chem Phys* (2017), Fig. 6)